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FRIDU: Functional Map Refinement with Guided Image Diffusion - Supplemental

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1. Implementation Details and Network Parameters

To enable training a diffusion model on a very limited dataset, we adopt the Patch-Diffusion paradigm from [WJZ*23] and build upon the code they provide. Specifically, we use their implementation based on the UNet-based diffusion model EDM-DDPM++ [KAAL22]. Following [WJZ*23], we employ the EDM sampling strategy with 50 deterministic reverse steps during inference.

In addition to default settings, we use the following hyperparameters for the Patch-Diffusion model: num_blocks $= 2$, model_channels $= 16$, channel_mult $= [2,4]$, and channel_mult_emb $= 0.1$. We train using 3 patch resolutions, where the functional maps have dimension 128×128 in all experiments—except in Section 4.2, where we match the resolution used in DiffZO which is 130×130. The parameter duration $= 2$ is used across all experiments, except in Section 4.2, where we set duration $= 10$ for the model trained on FAUST, and duration $= 15$ for the model trained on FAUST+SCAPE.

We use a batch size of 32 for all experiments, except for training on FAUST+SCAPE, where we use a batch size of 64. We run all experiments on a single NVIDIA A40 GPU, except for the FAUST+SCAPE training, which we run on NVIDIA L40S GPU.

For guidance, we implement the algorithm described in [BCS*23] with adaptation to the EDM.

2. Initial Functional Map Computation

We use the `pyFM` Python package [Mag21] to generate initial functional maps for the experiments in Section 4.1, which also includes the computation of WKS descriptors. For the WKS we use the parameters $n\_descr = 100$, $subsample\_step = 5$ based on 150 eigenvectors. For SHOT descriptors, we compute them in advance using the code from [HLR*19], with the parameters num_evecs $= 150$, num_bins $= 10$, and radius $= 15$.

3. DiffZO Comparison

For comparison with the DiffZo method [MO24], we extract the weights of their pretrained Feature Extractor model (DiffusionNet [SACO22]) for models trained on FAUST and FAUST+SCAPE. To compute the initial functional map used in our model, we first compute the pointwise map Π_{init} from the extracted features and then derive the corresponding functional map using the corresponding Laplace-Beltrami eigenbases.

4. Visualizations

For visualizing functional map matrices, we use the colormap defined in [LJO19], and for rendering in Blender, we utilize Blender-Toolbox [Liu18].

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